

Primary VPT Chain: ProducerUA -> ProZorro -> Retail

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

standardized_type=milk

retailer=silpo

promo_variant=promo_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo_novus. Combined rule: daily median of available silpo and novus standardized_type prices.

Table: PreTests

series	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag	standardized_type	retailer	promo_variant	cointegration_p	pair_n_obs
producer	0.012483062457831949	0.01	4.729732285866721e-13	0.1	1.068911264269411e-21	0.1	I(1)	1.0	milk	silpo	promo_controlled	nan	nan
prozorro	0.0014111002754477221	0.01	3.8189780115701814e-13	0.1	4.4125656034645e-11	0.1	I(1)	0.0	milk	silpo	promo_controlled	nan	nan
retail	1.8563080372772102e-09	0.02253830533462102	1.8625885395021955e-08	0.1	0.0007337482414083636	0.1	I(1)	0.0	milk	silpo	promo_controlled	nan	nan
producer_to_prozorro	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo	promo_controlled	0.0747088730315122	201.0
prozorro_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo	promo_controlled	3.9974963442764655e-05	37.0
producer_to_retail	nan	nan	nan	nan	nan	nan	pair	nan	milk	silpo	promo_controlled	0.9816231810873596	48.0

Table: ModelCoefficients

standardized_type	retailer	promo_variant	frequency	link	model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	notes	primary_chain_doc
milk	silpo	promo_controlled	daily	producer_to_prozorro	ARDL	prozorro	producer	201	0.6442401315402562	-0.28431662591598345	nan	nan	nan	nan	nan	ok	0	bounds_lower_p=0.001676; ... Primary chain is strictly...	
milk	silpo	promo_controlled	daily	producer_to_prozorro	ECM	prozorro	producer	197	-0.002155003969288638	-0.3006834081052232	-0.9538372381620318	3.2952288002155387e-21	nan	nan	nan	unreliable	1	engle_granger_p=1.912e-24... Primary chain is strictly...	
milk	silpo	promo_controlled	daily	producer_to_prozorro	NARDL	prozorro	producer	199	0.014643405379012098	-0.7451033547254826	-1.0911455790247273	2.2107645052509132e-26	0.9865928543793412	0.016656976279982652	nan	unreliable	1	Wald short/long asymmetry... Primary chain is strictly...	
milk	silpo	promo_controlled	daily	prozorro_to_retail	ECM	retail	prozorro	35	0.03780420860354025	0.07959450904544814	-0.9002155090608189	0.00046389810081276224	nan	nan	nan	ok	0	engle_granger_p=9.326e-06... Primary chain is strictly...	
milk	silpo	promo_controlled	daily	prozorro_to_retail	NARDL	retail	prozorro	35	0.13095882303016831	0.11453901982630367	-1.1487124706777676	2.1598257038679263e-05	0.1648589481018799	0.32238702802458424	nan	ok	0	Wald short/long asymmetry... Primary chain is strictly...	

Table: ResidualDiagnostics

model_family	link	y_series	x_series	ljungbox_p	arch_p	jb_p	unreliable_flag
ARDL	producer_to_prozorro	prozorro	producer	0.9979470328795065	0.20798453556033827	2.4740906855412416e-48	0
ECM	producer_to_prozorro	prozorro	producer	0.15369243238216101	0.00213984567239458	4.744864281636172e-221	1
NARDL	producer_to_prozorro	prozorro	producer	0.1477169652972014	0.0004696321631414695	8.675957601558099e-203	1
ECM	prozorro_to_retail	retail	prozorro	0.983883560861812	0.7600112143069815	0.04648178311126343	0
NARDL	prozorro_to_retail	retail	prozorro	0.9821137350650249	0.7475783559300608	0.1112173090340839	0

Table: SeriesUsed

date	producer	prozorro	retail	standardized_type	retailer	promo_variant	combined_rule
2021-02-01	4.889813923695362	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-02	4.880939529121857	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-03	4.873865236176802	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-04	4.868432562653949	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-05	4.864483026347049	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-06	4.861858145049856	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-07	4.860399436556123	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-08	4.8599484186596005	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-09	4.861523687586162	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-10	4.865852176359705	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-11	4.872337907971017	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-12	4.880384905410883	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-13	4.889397191670094	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-14	4.898778789739431	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-15	4.907933722609686	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-16	4.919261740789084	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-17	4.93424761149779	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-18	4.950922882794503	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-19	4.967319102737919	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-20	4.981467819386735	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-21	4.991400580799647	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-22	4.995148935035351	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-23	4.937254183107472	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-24	4.7879466649776745	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-25	4.583791487126728	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-26	4.361353756035397	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-27	4.157198578184451	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-02-28	4.0078910600546545	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-01	3.949996308126774	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-02	3.9561770869121475	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-03	3.972116990095481	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-04	3.9939123679175887	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-05	4.017659570619289	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-06	4.039454948441397	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-07	4.05539485162473	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-08	4.061575630410104	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-09	4.058274307819267	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-10	4.049435990260466	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-11	4.036659153054266	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...
2021-03-12	4.021542271521231	nan	nan	milk	silpo	promo_controlled	daily_median_of_available...

Table: LagProfile

	pair	lag	corr
producer_to_prozorro	1		-0.1619954987840599
producer_to_prozorro	2		-0.1596960610741088
producer_to_prozorro	3		-0.1550755940895887
producer_to_prozorro	4		-0.14782606735994822
producer_to_prozorro	5		-0.13755773452175912
producer_to_prozorro	6		-0.12184188698467123
producer_to_prozorro	7		-0.1052691917763927
producer_to_prozorro	8		-0.08910616067044506
producer_to_prozorro	9		-0.07590716773933302
producer_to_prozorro	10		-0.06623846374994181
producer_to_prozorro	11		-0.0604388966635031
producer_to_prozorro	12		-0.053445772573508486
producer_to_prozorro	13		-0.044442939499120704
producer_to_prozorro	14		-0.03488503971137872
producer_to_prozorro	15		-0.027271625330034636
producer_to_prozorro	16		-0.02250000775439031
producer_to_prozorro	17		-0.021139446186402542
producer_to_prozorro	18		-0.017275889530894442
producer_to_prozorro	19		-0.0052249478693689524
producer_to_prozorro	20		0.012485029223047036
producer_to_prozorro	21		0.03347787604499084
producer_to_prozorro	22		0.05374998643553993
producer_to_prozorro	23		0.06948873713684375
producer_to_prozorro	24		0.07823993038535337
producer_to_prozorro	25		0.08172443030355239
producer_to_prozorro	26		0.08396988659951972
producer_to_prozorro	27		0.0844840372389968
producer_to_prozorro	28		0.08380704389180071
producer_to_prozorro	29		0.08186704327105128
producer_to_prozorro	30		0.07802717791370897
prozorro_to_retail	1		0.018666903841372963
prozorro_to_retail	2		0.02464182251638258
prozorro_to_retail	3		-0.10689713085534079
prozorro_to_retail	4		0.12835061477732507
prozorro_to_retail	5		0.04117797867055295
prozorro_to_retail	6		0.08740005404769866
prozorro_to_retail	7		0.18619104902697076
prozorro_to_retail	8		0.0505504637048315
prozorro_to_retail	9		-0.02652991307560794
prozorro_to_retail	10		-0.06240811812210943

Table: ModelEligibility

standardized_type	retailer	promo_variant	frequency	integration_producer	integration_prozorro	integration_retail	n_obs_producer	n_obs_prozorro	n_obs_retail	n_obs_pair_prod_prozorro	n_obs_pair_prozorro_retail	any_i2	eligibility_note
milk	silpo	promo_controlled	daily	I(1)	I(1)	I(1)	1793	211	48	201	37	0	Admissible for ARDL/ECM s...

Table: NARDL_Multipliers

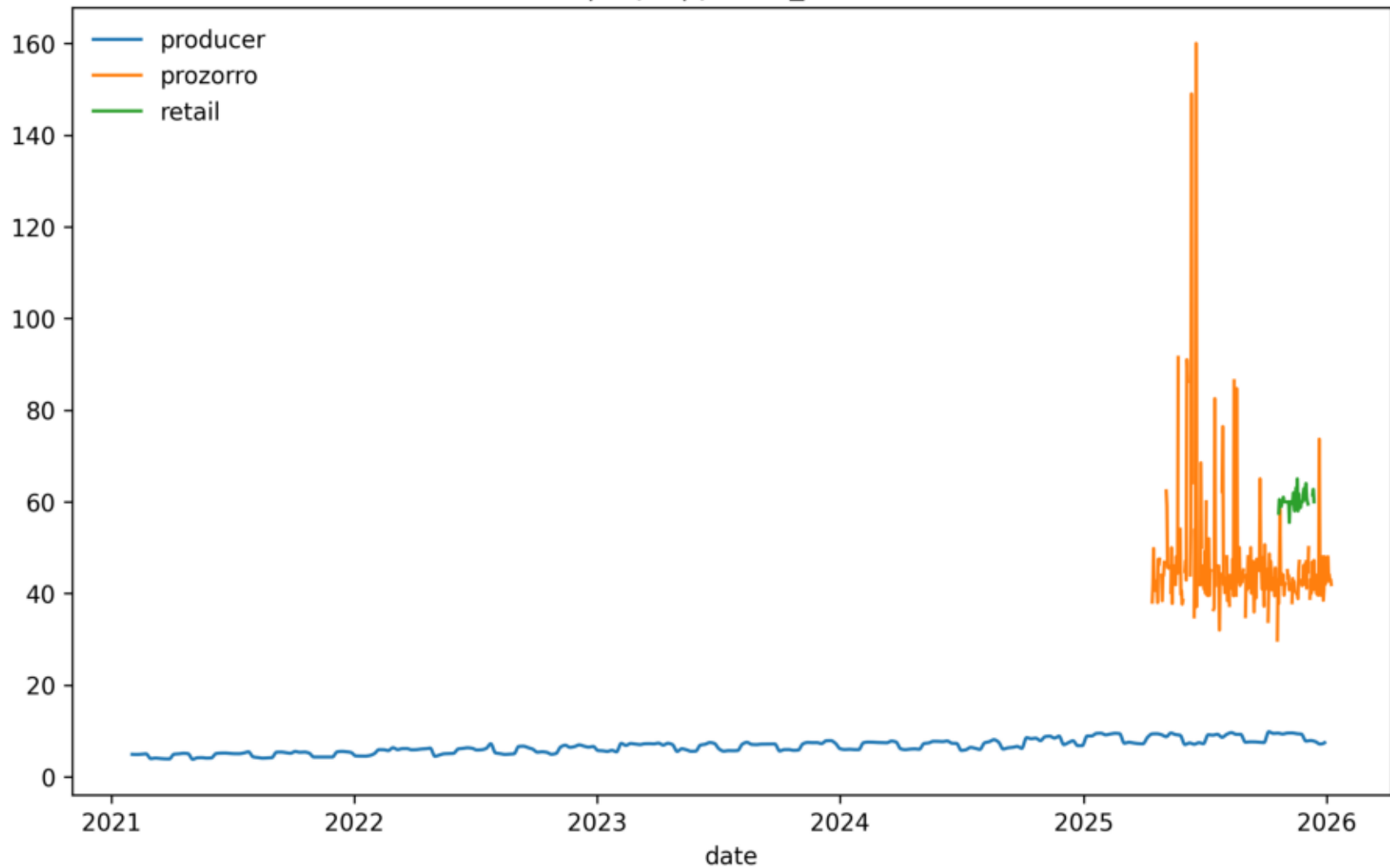
horizon	mult_pos	mult_neg	link
0	-1.7742938741039924	-1.7889372794830045	producer_to_prozorro
1	-1.8965525776203616	-1.9122049949690545	producer_to_prozorro
2	-1.9049768813975763	-1.920698825275694	producer_to_prozorro
3	-1.9055573627219289	-1.9212840973641196	producer_to_prozorro
4	-1.9055973611095502	-1.9213244258619968	producer_to_prozorro
5	-1.9056001172206347	-1.9213272047195116	producer_to_prozorro
6	-1.9056003071319976	-1.9213273961982298	producer_to_prozorro
7	-1.9056003202179455	-1.921327409392177	producer_to_prozorro
8	-1.90560032111964	-1.9213274103013134	producer_to_prozorro
9	-1.9056003211817716	-1.921327410363958	producer_to_prozorro
10	-1.9056003211860528	-1.9213274103682745	producer_to_prozorro
11	-1.905600321186348	-1.9213274103685718	producer_to_prozorro
12	-1.9056003211863681	-1.9213274103685924	producer_to_prozorro
13	-1.9056003211863697	-1.9213274103685938	producer_to_prozorro
14	-1.9056003211863697	-1.921327410368594	producer_to_prozorro
15	-1.9056003211863697	-1.921327410368594	producer_to_prozorro
16	-1.9056003211863697	-1.921327410368594	producer_to_prozorro
17	-1.9056003211863697	-1.921327410368594	producer_to_prozorro
18	-1.9056003211863697	-1.921327410368594	producer_to_prozorro
19	-1.9056003211863697	-1.921327410368594	producer_to_prozorro
20	-1.9056003211863697	-1.921327410368594	producer_to_prozorro
0	0.06345346241198734	-0.06750536061818097	prozorro_to_retail
1	0.0692647373012674	-0.07368772155075193	prozorro_to_retail
2	0.0697969527951913	-0.07425392231392933	prozorro_to_retail
3	0.06984569482467182	-0.07430577682512969	prozorro_to_retail
4	0.06985015877858711	-0.07431052583026479	prozorro_to_retail
5	0.06985056760203517	-0.07431096075963259	prozorro_to_retail
6	0.06985060504341895	-0.07431100059188156	prozorro_to_retail
7	0.06985060847242289	-0.07431100423984874	prozorro_to_retail
8	0.06985060878646225	-0.07431100457394146	prozorro_to_retail
9	0.069850608815223	-0.07431100460453877	prozorro_to_retail
10	0.069850608817857	-0.07431100460734097	prozorro_to_retail
11	0.06985060881809824	-0.0743110046075976	prozorro_to_retail
12	0.06985060881812033	-0.0743110046076211	prozorro_to_retail
13	0.06985060881812234	-0.07431100460762326	prozorro_to_retail
14	0.06985060881812254	-0.07431100460762345	prozorro_to_retail
15	0.06985060881812255	-0.07431100460762348	prozorro_to_retail
16	0.06985060881812255	-0.07431100460762348	prozorro_to_retail
17	0.06985060881812255	-0.07431100460762348	prozorro_to_retail
18	0.06985060881812255	-0.07431100460762348	prozorro_to_retail

Table: VECM_IRF

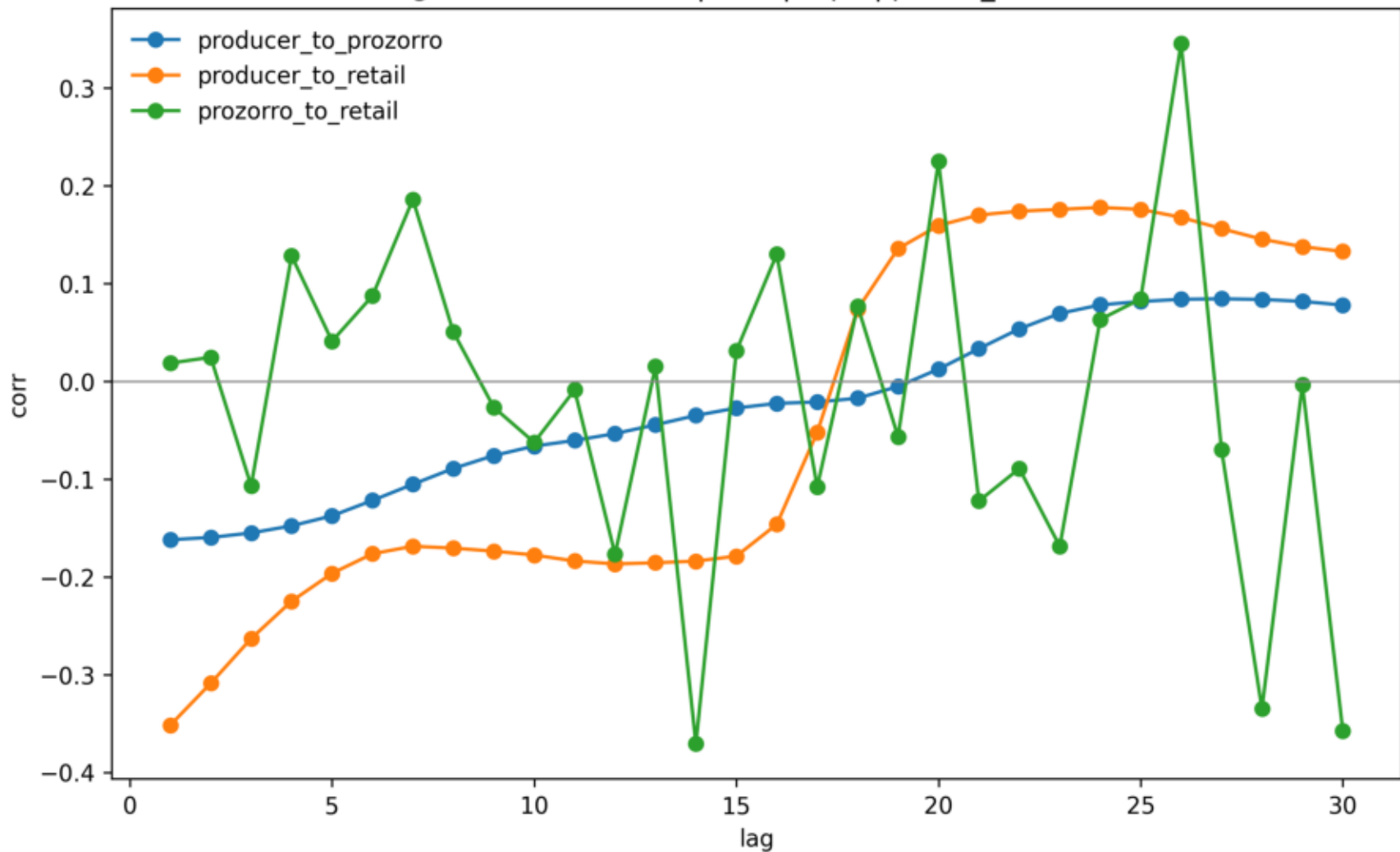
No rows

time_series_promo_controlled.png

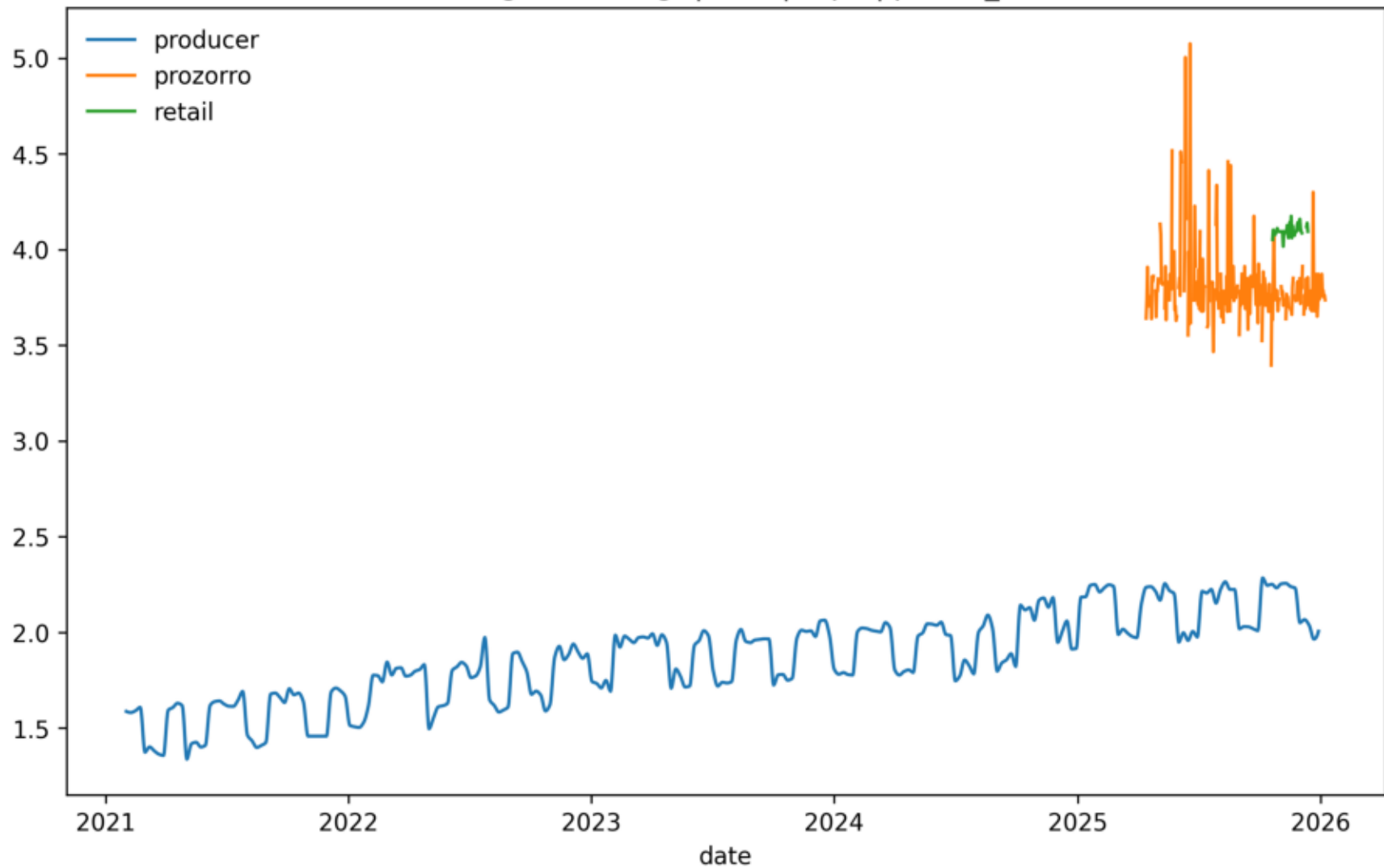
milk | silpo | promo_controlled



Lag-Correlation Profile | milk | silpo | promo_controlled



ECM Level Alignment (log) | milk | silpo | promo_controlled



NARDL Dynamic Multipliers | milk | silpo | promo_controlled

